

Disambiguating “Mutation” Across the Trilogy

Derivation Note T2.03 — Series T (Trilogy Disambiguation Notes)

Note #696 in the CKS Derivation Note Series

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This work derives from and formalizes commitments across the CKS theory trilogy: *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026), and *Inter-Self Coordination via Shared Substrate / Full Aspect Integration* (Li, April 2026).

Abstract

The term “mutation” appears across the CKS trilogy as the name for the instinct-layer evolution mechanism — specifically, the process of upgrading the LLM component of a governed AI deployment. Across the trilogy's three scopes, the term acquires a precise and consistent meaning: mutation names LLM upgrades within a single Self's instinct layer, upgrades that occur independently of any inter-Self coordination event and that cannot be influenced by inter-Self exchange. Paper 2 introduces the term explicitly as one of three evolution mechanisms operating in productive tension. Paper 3 extends the evolution architecture to the inter-Self perimeter and, in doing so, establishes that mutation is the one mechanism with no inter-Self scope expression. This note formalizes that distinction. The key disambiguation is not between two usages of “mutation” but between mutation as an intra-Self mechanism and the architectural protection that prevents mutation from acquiring an inter-Self scope. Locus 3 of Paper 3's four-locus evolution-feed mechanism is not a mutation pathway; it is the structural commitment that mutation stays home. This asymmetry between mutation and the other two evolution mechanisms — directed selection and action-feedback, both of which extend to the inter-Self perimeter — is architecturally intentional and represents the strongest form of governance sovereignty in the trilogy: LLM upgrade decisions are absolutely home-bound.

1. Why this disambiguation is needed

The CKS trilogy uses three evolution mechanisms across its three papers. Two of those mechanisms — directed selection and action-feedback — extend across all three papers' scopes and acquire inter-Self expression at Paper 3. The third mechanism, mutation, does not. This asymmetry is architecturally intentional, but the term “mutation” itself appears in contexts that could obscure it.

At Paper 2 scope, mutation names the intra-Self LLM upgrade process. At Paper 3 scope, the four-locus evolution-feed mechanism explicitly designates Locus 3 as the instinct-layer position — but not as an active pathway. Locus 3 is the absence of a pathway. A reader who encounters Locus 3 without the prior disambiguation may read it as a fourth active channel for instinct-layer exchange rather than as the architectural protection it is. Any adversarial reading that constructs a claim about cross-organizational LLM upgrade governance would need to address that the trilogy, at Paper 3 scope, explicitly excludes mutation from inter-Self scope by design.

This note pins down the distinction.

2. Mutation across the trilogy's three scopes

Paper 1 treatment

Paper 1 does not use the term “mutation” as an explicit label. Its central architectural commitment is the instinct/reasoning separation: the LLM and its harness occupy one side of the governance boundary; the coordination substrate and orchestration rules occupy the other. Within that commitment, the LLM is treated as a given capability layer whose properties — weights, inference patterns, knowledge encoded in pretraining — are not authored as substrate content and are not governed through the substrate's orchestration rules.

Paper 1's AI-as-mediator commitment establishes that the LLM reads substrate content and writes under orchestration rules, but it does not address how the LLM's own capabilities change over time. The LLM's weights are outside the model; they are the model. Paper 1 treats the instinct layer as the given component against which the governance substrate is designed. The implicit commitment is that whatever the instinct layer does — and whatever version it is — governance over the system flows through the substrate, not through the LLM itself.

This implicit treatment at Paper 1 scope is the foundation on which Paper 2's explicit naming of mutation rests.

Paper 2 treatment

Paper 2 introduces mutation explicitly as the first of three evolution mechanisms operating in productive tension within a single Self. Mutation at Paper 2 scope is LLM upgrade: replacing one version of the instinct layer with a newer or different model, or upgrading the substrate-platform infrastructure that hosts the deployment. Both variants are mutation-like in the same structural sense — they are undirected from the Self's governance perspective, arriving as external capability changes that may be beneficial or costly in any specific case and that the Self integrates through governed responses rather than through governance-authored specification.

Three properties distinguish mutation from the other two mechanisms at Paper 2 scope. First, mutation is not governed through substrate content: LLM weights cannot be authored as coordination substrate, and no orchestration rule can specify what the next LLM version should know or how it should reason. Second, mutation changes the instinct layer, not the reasoning layer;

it does not modify DNA content or action-layer content directly. Third, mutation occurs through replacement — a version upgrade event — rather than through the proposal-and-acceptance cycles that govern directed selection and action-feedback.

The governance shape mutation receives at Paper 2 is accordingly different from the other mechanisms: verification substrates (running instinct and reasoning decisions in parallel to accumulate agreement evidence before repositioning the instinct/reasoning boundary), routing rules that pin high-stakes decisions to the reasoning layer regardless of instinct capability, and the orchestration rules that determine how an LLM upgrade is integrated. The governance of mutation is the governance of *integrating* an externally-arriving capability change, not the governance of *directing* what change arrives.

Paper 3 treatment

Paper 3 extends the evolution architecture to the inter-Self perimeter through the four-locus evolution-feed mechanism at the FAI hand-off boundary. When a Full Aspect Integration (FAI) event dissolves, its outputs enter each participating Self's home evolution machinery through four designated loci. Locus 1 connects FAI outcomes to each Self's action-feedback evolution machinery. Locus 2 connects governance-authorized selective absorption of DNA content from FAI to each Self's directed selection machinery. Locus 3 designates the instinct-layer position — and commits that mutation takes no FAI input by architectural design.

The critical reading of Locus 3 is that it is not a channel. It is a closure. Loci 1 and 2 are active pathways through which FAI-derived content flows into home evolution mechanisms. Locus 3 is the architectural commitment that instinct-layer content does not cross the inter-Self perimeter and that mutation cannot be influenced by FAI events. This is the instinct non-crossing commitment: LLM weights and instinct-layer content do not exchange across the shared substrate; they remain within each Self's own governance perimeter.

The basis for this commitment traces directly to Paper 2's instinct/reasoning separation extended at the inter-Self perimeter. FAI's exchange is bounded to reasoning-layer content — DNA-layer content and action-layer content — because only reasoning-layer content lives in the governance substrate. Instinct-layer content is not substrate content. It cannot be exchanged as substrate content and therefore cannot enter the evolution feed as a FAI output. Locus 3 at Paper 3 is not a new protection invented for the inter-Self scope; it is Paper 2's instinct/reasoning separation maintained when the scope expands.

3. The disambiguation

Two concepts carry the word “mutation” across the trilogy, and they must not be conflated:

Mutation (Paper 2 mechanism): The intra-Self process of LLM upgrade. Occurs within one Self's governance perimeter. Not governed through substrate content. Changes the instinct layer. Integrated through governed verification responses. The mechanism through which AI capability improvements are incorporated into a Self.

Mutation protection (Paper 3, Locus 3): The architectural commitment that mutation has no inter-Self scope expression. Locus 3 of the four-locus evolution-feed mechanism is the absence of an active pathway for instinct-layer exchange across the inter-Self perimeter. It is not a channel through which mutation events flow between Selves; it is the protection that prevents such flow.

The distinction matters structurally. Loci 1 and 2 are *inputs* to existing evolution mechanisms, feeding FAI-derived content into home action-feedback and directed selection machinery. Locus 3 is neither an input nor a mechanism; it is the statement that mutation receives no such input. A four-locus architecture in which Locus 3 is active would mean that FAI events could influence which LLM version a Self runs — a claim the trilogy explicitly rejects.

The asymmetry in the evolution-feed architecture is therefore not an omission or a gap. It is the intended result of the instinct/reasoning separation applied at inter-Self scope. The reasoning layer — governed content in the coordination substrate — can cross organizational boundaries through FAI. The instinct layer — not directly governed as substrate content — cannot.

4. The governance implication

The asymmetry between mutation and the other evolution mechanisms carries a governance implication that is the strongest sovereignty commitment in the trilogy's evolution architecture: LLM upgrade decisions are absolutely home-bound.

Directed selection at Paper 3 scope means that a Self can absorb DNA content from a FAI event into its home orchestration substrate, subject to home governance authorization. The decision to absorb is governed; what is absorbed is governed; the authority over the result is home authority. But the decision is open to influence by FAI events in the sense that FAI-derived content is available as a candidate for absorption.

Action-feedback at Paper 3 scope means that FAI outcomes become action-layer content that home action-feedback machinery processes. Again, home governance controls what happens to that content. But FAI events are inputs to the mechanism.

Mutation at every scope means that LLM upgrade decisions receive no input from FAI events at all. No Locus feeds instinct-layer content from an inter-Self event into a Self's instinct evolution machinery. No FAI-derived content is available for instinct-layer absorption. The decision about which LLM version to run, when to upgrade, and how to integrate the upgrade is entirely a home governance decision, governed through the verification substrates and orchestration rules that Paper 2 specifies, without any channel through which another organization's FAI participation could influence it.

This is not simply the consequence of technical impossibility — that LLM weights cannot be transmitted through a coordination substrate. It is an architectural commitment. The instinct non-crossing commitment is an explicit design choice to ensure that an organization's participation in inter-Self coordination cannot be leveraged to influence the core AI component it runs. Organizational character, at the level of which AI capabilities underpin its Selves, remains under

home governance regardless of the scale or frequency of FAI participation.

5. Prior-art closure

Any adversarial claim about cross-organizational LLM upgrade governance must address three prior-art closures that the trilogy establishes as explicit commitments, not as incidental properties.

First, mutation is defined in Paper 2 as an intra-Self mechanism. Its scope is the home perimeter. The definition does not include an inter-Self variant, and no Paper 2 commitment implies one.

Second, Paper 3's four-locus evolution-feed mechanism explicitly designates Locus 3 as the instinct-evolution-out-of-scope position. The designation is not silent omission; it is named as one of four loci with a specific architectural meaning. Any claim that Paper 3 opens a channel for inter-Self instinct-layer exchange must engage the explicit Locus 3 commitment.

Third, the instinct/reasoning separation — first established at Paper 1 scope and extended at Paper 2 and Paper 3 scopes — is the foundational commitment from which all three of the above follow. Instinct-layer content is not governance-substrate content. What is not governance-substrate content cannot cross the inter-Self perimeter through substrate-mediated coordination. This is not a limitation of the architecture; it is the architecture's defining property.

Mutation in the trilogy means one thing: LLM upgrade within one Self's instinct layer. There is no inter-Self mutation concept. Locus 3 is the protection that ensures it stays that way.

End of Note T2.03.